

UTTARANCHAL UNIVERSITY

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PRACTICAL → 1

Q Execute the following network oriented commands (with all their options) and observe their output

- PING
- TRACERT
- ROUTE
- IPCONFIG
- ARP
- NETSTAT
- NBTSTAT
- HOSTNAME
- NETSEND
- DNSCONFIG

* IPCONFIG :- It provides essential details such as IP address, subnet mask, default gateway for all active adapters.

* IPCONFIG/all :- Users can view detailed information including MAC address, DNS servers, and DHCP status.

* nslookup (Name Server Lookup) :- This command used to map domain name with its IP address. It is available on windows, LINUX, MAC, OS, operating system.

* Ping :- Ping Command checks ~~whether~~^{whether} a computer on a network on the internet is reachable and measures the round-trip time of message sent from your system to the target.

* Tracert :- Tracert Command is a network tool used in windows to determine the path that data packets take from your computer to destination computer. ~~that~~ It shows each hop along the way and how long it takes for the packet to travel through hop.

- * hostname:- displays name of the computer
- * netstat:- Netstat command displays active network connection, routing table and various interface statistics.
- * net send:- It was a utility in older version of windows (like windows NT, 2000, XP, server 2003) used to send messages to other users, computers, It work with the messenger service, which has been removed in later windows
- * DNS Configuration:- (Domain name System)

Handwritten signature and date:
21/8/28

PRACTICAL → 2

9. Design LAN Cable - straight LAN cable and Cross LAN Cable.
Cisco Packet Tracer is a powerful network simulation and visualization tool developed by Cisco System. It is used for learning and practicing networking concepts without the need for physical devices.

Straight Through Cable →

A type of ethernet cable where the wires on both ends are arranged in the same order. It connects both ends, either (T568B - T568B)

Connects similar ~~for~~ types of devices.

Ex → PC ↔ Switch

Cross Over Cable →

A type of ethernet cable where the wires on one end are reversed.

It connects similar device directly.

Ex → PC ↔ PC

Switch ↔ Switch.

Router ↔ Router.

Hub ↔ Hub

Lakshmi
28/8/28

PRACTICAL - 03

Q Create a Network of Two Computers A and B through Creating LAN Cable in Practice - 02, check connectivity & test them with the help of PING command.

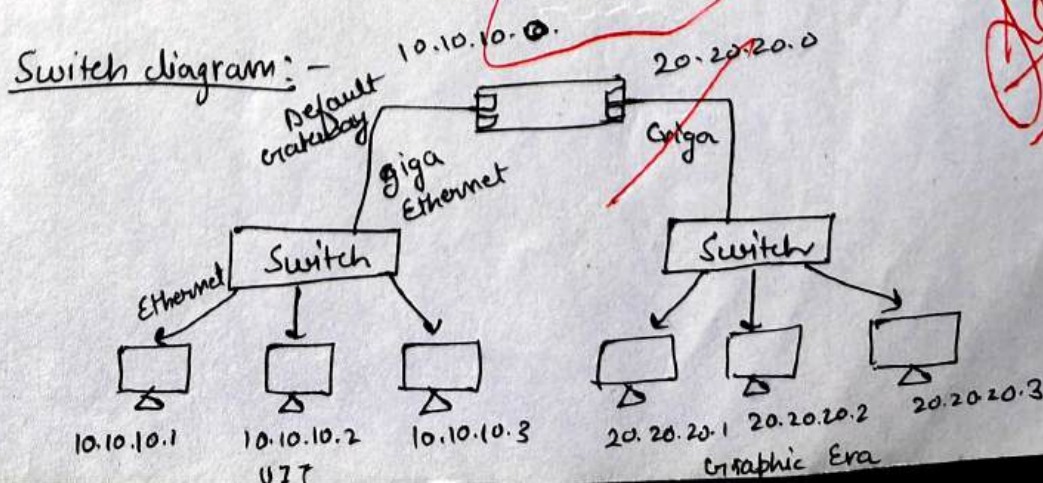
Ans . Switch devices : It is a intelligent device use MAC cable for communication

- One-to-One Communication:- Two cables straight and cross element cable are utilize for designning a network using switch devices
- It is used for Intra-Network.
- Each switch devices has ethernet port and giga ethernet Port.

Router:- Responsible for Network-to-Network device
~~contain~~ • Intelligent devices contain routing information

- Use default gateway i/p address for making a communication
- It contains gigaethernet ports.

Repeater:- Used for long distance communication to reboot or amplify quick signal so that signal can be reached without any noise and error.



Practical + 4

Design networks using BUS, STAR, TREE, RING topology for N nodes and check their connectivity through the PING command whether they are connected or not

BUS Topology :-

- Single backbone cable.

- All nodes are connected to the main cable.
- If backbone fails, whole network fails

Star

Topology :-

- All nodes are connected to a central hub.
- Easy to maintain and widely used.

Ring

Topology -

- Each node is connected to exactly 2 other nodes forming a closing loop.

- Data travels in one direction until it reaches the destination.

Tree Topology -

- Combination of star and bus topology

- Hierarchical structure with root node and branches.

Ping Command -

- It is used to test the connectivity b/w two devices

Lakshmi
30/10/25

Practical - 5

Implementation of data link layer framing methods
character stuffing and bit stuffing.

Introduction - Two common methods for framing data of data link layer are character stuffing and bit stuffing.

- This method ensures that data sent across the network is properly framed.
- Special character or bit pattern within the data.

Character stuffing - It involves inserting special characters into data to frame boundary this method used text data.

- Flag: denote beginning and end of frame.
- Escape: Special character used to escape the flag character within the data.

Bit Stuffing :- It involves inserting non-info bits into data to break up patterns.

- Flag Sequence: A special sequence of bits to denote beginning and end of the frame.
- Stuffed bit is bit inserted to ensure that flag sequence does not appear within the data.

Each frame begins and ends with special bit pattern called flag byte.

Lakshmi

Program Code → // BIT Stuffing program

```
#include <stdio.h>
#include <string.h>
```

```
void main()
```

```
{
    int a[20], b[30], i, j, k, count, n;
```

```
printf("Enter frame length:");
```

```
scanf("%d", &n);
```

```
printf("Enter input frame (0's & 1's only):");
```

```
for(i=0; i<n; i++)
```

```
scanf("%d", &a[i]);
```

```
i=0; count=1; j=0;
```

```
while(i<n)
```

```
{
```

```
if(a[i] == 1)
```

```
{
```

```
b[j] = a[i];
```

```
for(k=i+1; a[k] == 1 && k<n && count < 5; k++)
```

```
{
```

```
j++;
```

```
b[j] = a[k];
```

```
count++;
```

```
if(count == 5)
```

```
{
```

```
j++;
```

```
b[j] = 0;
```

```
}
```

```
i=k;
```

```
}
```

```
else
```

```
{
```

```
b[j] = a[i];
```



```

}
i++;
j++;
}
printf("After stuffing the frame is:");
for(i=0; i<j; i++)
printf(" %d", b[i]);
}

```

Program Output:

Enter frame Length:

Enter input frame (0's & 1's only):

1
1
1
1
1

After stuffing the frame is: 111110

(Program exited with code: 6)

Safar

Practical - 06

Configuring a Router

Objective - Configure Router in LINUX with Packet Tracer
Cisco simulator.

Requirement - Command and Packet Tracer.

Theory: Introduction - In this practical, we will explore how to configure a router using Cisco Packet Tracer, a powerful network simulation tool, and perform basic routing configuration that could be mirrored in a LINUX environment. Packet Tracer allows for design, configuration, and Troubleshooting of network step in a virtual environment, making it an ideal tool for learning and experimentation.

Understanding Router Configuration:-

Routers are critical devices in networking that forward data packets b/w computer networks. They perform traffic directing functions on the Internet and are essential for connecting different subnets.

Basic Router Configuration Tasks →

1. Assigning IP addresses to interfaces.
2. Configuring routing protocols.
3. Setting up static routes.
4. Verifying configuration and connectivity.

Setting up the Network in Packet Tracer →

Step 1: Launch Packet Tracer and create a New Project.

1. Open Cisco Packet Tracer.
2. Create New Project by clicking on File > New.

Step 2: Add Devices to workspace,

1. Add a Router:

• In the device list, select Routers.

• Drag and drop a router (eg. 1941) into the workspace.

2. Add Computers (Nodes):
 - In a device list, select End Services.
 - Drag and drop 2 computers (eg. PC - PT) into the workspace.
3. Add a Switch:
 - In the device list, select Switches.
 - Drag & drop a switch (eg. 2960) into the workspace.

Step 3: Connect the Devices

1. Connect the Router to the switch:
 - select the Connections tool from the toolbar.
 - Choose Copper Straight-Through cable.
 - Connect GigabitEthernet 0/0 on the router to any port on the switch.
2. Connect the Computers to the switch:
 - Connect each computer to the switch using Copper Straight-Through cables.

Step 4: Configure IP Addresses.

Assign IP addresses to the computer and router interfaces.

On Computers -

1. Click on computer to open its configuration window.
2. Navigate to the Desktop tab and select IP configuration.
3. Assign IP address as follows:

- PC 1: Address: 192.168.1.2 Subnet mask: 255.255.255.0
- PC 2: Address: 192.168.1.3 Subnet mask: 255.255.255.0

On the Router -

1. Click on the Router to open its configuration window.
2. Navigate to the CLI tab to access the Command Line Interface.

Enter the following Commands to configure the router:

plaintext

Copy Code

Router> enable

Router # configure terminal

Router (config) # interface GigabitEthernet 0/0

Router (config-if) # ip address 192.168.1.1 255.255.255.0

Router (config-if) # no shutdown

Router (config-if) # exit

Router (config) # exit

Router # copy running-config startup-config

Verifying Configuration and Connectivity.

Step 1: Check Interface Status.

On the router CLI, use the following command to check the status of the interface:

plaintext

Copy code

Router # show ip interface brief

Ensure the GigabitEthernet 0/0 interface shows the correct IP address and status as up.

Step 2: Test connectivity with Ping

1. open the Command Prompt on PC1.

2. Ping the Router's interface:

ping 192.168.1.1

3. Ensure the ping is successful.

Repeat the process on PC2.

Step 3: Test connectivity b/w PCs.

1. open the Command prompt on PC1.

2. Ping PC2:

plaintext

Copy Code

ping 192.168.1.3

3. Ensure the Ping is successful.

Lakshmi
30/10/28

Practical - 07

Objective:- Configure switch in LINUX with Packet Tracer Cisco simulator.

Requirement:- Command Prompt and Packet Tracer.

Theory:-

Introduction:- In this practical, we will explore a process of configuring a switch in a LINUX environment using Cisco's Packet Tracer. It is a powerful network simulation tool that allows user to create network topologies and emulate modern computer networks. It is an essential tool for anyone looking to learn about network design, configuration, and troubleshooting.

Step 1:-

- Add one Cisco Router, two Cisco Switches, and 4 PC's to the program's workspace. Then note the port and slot numbers of the Router's GigabitEthernet interfaces in the workspace.
- Double click on the Router to open the IOS command Interface.

Step 2:-

- In the Cisco Router CLI ~~view~~ window, assign an IP address to the GigabitEthernet 0/0 interface and execute the no shutdown command to activate the interface.

Step 3:-

- After configuring the GigabitEthernet 0/0 interface, the port will be green. The meaning of this colour is that the interface is active and working.

Step 4: To assign an IP address to the GigabitEthernet 0/2 interfaces and activate the port, perform the following commands.

Step 5:-

- In the following image, you can see that the Router's GigabitEthernet ports are connected to the switches and are active.

Step 6:-

- Now, we need to assign an IP address to the Computers that you add to the topology. To do this, click on the PC to which you will assign an IP, and on the Desktop tab, click IP configuration.
- Then, it would be best if you addressed PC according to the IP block of the network on which the network is. According to the topology above, assign an IP from block 192.168.1.0/24 as PC0 is connected to the network under the Router's Gig 0/0 interface.
- Configure IP address, subnet mask, and default gateway settings for PC0 as in the following image.

Step 7:-

- Click Command Prompt on PC0, and then Ping the Cisco Router Gig 0/0 default gateway to test the network connection.

Step 8:-

- This time, the TCP/IP setting of PC3 on the 192.168.2.0/24 network are configured as follows.

Step 9:-

- When we ping gig 0/1 interface (default gateway) via PC3, we can also see that operations was successful.

Step 10:-

- Likewise, suppose you ping Router's gig 0/2 interface via PC0. In that case, the operation will still be successful because on Router's main task is to communicate with each other the different networks connected to it.

Step 11:-

- Ping operation to PC2 computer with IP address 192.168.2.10 via PC0.

Step 12 :-

- Ping operation to PC2 computer with IP address 192.168.2.20 via PC0;

Step 13 :-

- You can examine the MAC addresses of computers when you execute the show arp command at the Cisco router CLI command prompt.

Hafshi
20/10/28

Practicle - 09

Simulation of wired and wireless network using NS2/NS3 simulators.

ABSTRACT

In Computer terminology the definition of network is similar as a group of a computer logically connected for the sharing information or services.

General Terms

Network, Routing, wireless Protocol, NS2 simulator

Keywords

Wired Network, wireless Network, ZRP, TORA, NS2

1. Introduction

A network is defined as the group of people or systems or organisation who tends to share their information collectively for their business purpose.

1.1 Wired Network

Wired Network, also called Ethernet network, are the most common type of local area Network technology. A wired network is simply a collection of two or more computers, printers, and other device linked by Ethernet cables.

1.2. Wireless Network

An ad-hoc network is a collection of wireless mobile hosts forming a temporary network without the aid of any stand-alone infrastructure or centralized administration.

2. Technique for N/w Performance Evaluation -

Performance is a key criterion in the design, procurement and use of computer system. Computer system professionals such as computer system engineers, scientist, analysts and users need to gain knowledge of performance for a given evaluation techniques as the goal to get the highest performance for a given cost.

2.1 Computer Network Simulation Tools

There are many simulators such as network simulators 2 (NS-2), OPNET Model, Glomaxim, OMNet++, and etc. In this project we choose Network Simulations tools NS2.

2.2 Advantages and Disadvantages of NS2

Advantages -

- cheap - Does not require costly equipment.
- Complex Scenarios can be easily tested.
- Support Protocols.
- Modularity
- Popular.

Disadvantages -

- Real system too complex to model. i.e. Complicated structure.
- Bugs are unreliable.
- Traffic scenario for wired and wireless Net..

3. Traffic Scenario for wired and wireless Network

3.1 Traffic scenario for wired net using TCL script

3.1.1 Static wired Network :- In this routing if the link is break b/w nodes then routing is Not possible. But when a link then there is no routing is generated. Packet transfer failed from one node to another.

3.1.2 Dynamic wired Network :- In this if the link breaks b/w nodes then routing is possible from other routes.
→ Distance Vector Routing Protocol is used used when we generated a Dynamic wired net, on simulation traffic follows one node to desired destination.

3.2 Traffic Scenario for wireless network using TCL Script

In this network we create a wireless network traffic scenario by TCL script.

- four nodes
- DSDV Routing Protocol.
- Data Transfer b/w nodes 1 to 3 using UDP.
- TCP, UDP agents
- CBR, FTP Application.

In Mobile Ad-hoc Nws are self-organizing and self-configuring multi-hop wireless Nws where, the structure of Nws changes dynamically.

4. Performance Analysis of ZRP and TORA

In wireless network we used so many routing protocols. In this paper we show how we can calculate the performance of routing protocols. ZRP and TORA are evaluated based on two performance matrices which are.

1. Packet delivery function
2. Average end to end delay.

5. Result and Discussion

5.1 Comparison of TORA and ZRP with graphs

According to graph TORA & ZRP on the base of packet delivery function, ZRP has some constant rate for packet delivery as compared to TORA.

5.2 Discussion

In this paper firstly we analysed the wired and wireless network and found that in static routing, if the link is break b/w nodes then routing is not possible. In this routing scheme there is no routing protocols used. In Dynamic Routing if the link breaks b/w nodes then routing is possible from other routes. Distance Vector Routing Protocols is used in Dynamic Routing.